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MASTER THESIS IN MATHEMATICS
6. JANUARY 2024



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Chapter 1

We gotta start somewhere

In this thesis we will be studying the Vicsek set, so therefore we will in this section start by constructing the set and proving some basic properties of the Vicsek set.

1.1 Construction the Vicsek set

Let $V_0 = \{p_1, p_2, p_3, p_4, p_5\}$ be the set of vertices in a plus with a middle point in $\mathbb{C}$. Then define

$$\mathcal{S}_i(z) = \frac{1}{3}(z - p_i) + p_i, \quad \text{for } i = 1, 2, \dots, 5.$$

The Vicsek set K is then the unique non-empty compact subset in $\mathbb{C}$ such that

$$K = \bigcup_{i=1}^5 \mathcal{S}_i(K)$$

V_0 is the boundary of K .

Next we can then define a sequence of sets $(V_m)_{m \geq 0}$ inductively:

$$V_{m+1} = \bigcup_{i=1}^5 \mathcal{S}_i(V_m)$$

this gives us a sequence of Vicsek set graphs $(G_m)_{m \geq 0}$ whose vertices are $(V_m)_{m \geq 0}$. Further we will use the notation of $V_* = \cup_{m \geq 0} V_m$ and for any $p \in V_m$ we denote the collection of neighbours of p in G_m by $V_{m,p}$. Lastly note that $\#V_m = 5^m \cdot 4 + 1$

1.2 Minimizing Energy

In this section we will follow arguments given by [Str06].

Given a finite connected graph K and it vertices V_m the for a function $u : V_m \rightarrow \mathbb{R}$ we define the energy to be

$$E_K(u) = \bigotimes_{x \sim y} (u(x) - u(y))^2$$

Remark: While [Str06] often calls this form graph energies, it is the Dirichlet form.

Proposition 1.1: Equation (1.2) is indeed a Dirichlet form

Proof. obvious? □

Given some graph K and a function $u^{(m)} : V_m \rightarrow \mathbb{R}$, let's say we want to extend this function to V_{m+1} . This leads us to defining an extension in the following way

Definition 1.2: Let K be a graph and $u : V_m \rightarrow \mathbb{R}$, then an *extension* of u is a function $u' : V_{m+n} \rightarrow \mathbb{R}$ where $u' \upharpoonright_{V_m} = u$.

It is obvious that $E_{m+n}(u') \geq E_m(u)$. Further one quickly notices that there must be a function that minimizes the energy of a given graph. We denote such a function with $\tilde{u}$ and call it a *harmonic extension*.

Definition 1.3: Let $\tilde{u} : V_{m+n} \rightarrow \mathbb{R}$ with $|_{V_m} = u$ is called a *harmonic extension* if $E_{m+n}(\tilde{u}) \leq E_{m+n}(u')$ for all $u' : V_{m+n} \rightarrow \mathbb{R}$ with $u'|_{V_m} = u$.

In some cases we have that $E_{m+n}(\tilde{u}) = rE_m(u)$ for some $r \in (0, 1)$. But instead of normalizing the values on E_m we will normalize E_{m+n} , and getting

$$r^{-1}E_{m+n}(\tilde{u}) = E_m(u), \quad \text{and} \quad r^{-1}E_{m+n}(u') \geq E_m(u)$$

for every extension u' of u . From the equality above we can define the following

Definition 1.4: The *renormalized graph energies* are defined as

$$\mathcal{E}_m(u) = r^{-m}E_m(u)$$

Remark 1.5: The sequence $\{\mathcal{E}_m(u)\}$ is a non-decreasing sequence. Something with it being constant if u is linear and how is u defined? In the sequence $\{\mathcal{E}_m(u)\}$ if we have u defined on V_m what is $\mathcal{E}_{m+1}(u)$? Any extension or the harmonic extension?

Now that we have 1.4, we can define what it means to be a harmonic function.

Definition 1.6: u is a *harmonic function* if u minimizes $\mathcal{E}_m(u)$ for all m , given boundary values V_0 .

Minimizing Energy on the Vicsek set

In the previous section we have talked about a harmonic function, but without every constructing one. We will therefore in this next section show how to create an harmonic function for the Vicsek set for any start boundary values.

Given $V_0 = \{q_0, q_1, q_2, q_3, q_4\}$ and $u : V_0 \rightarrow \mathbb{R}$ with $u(q_i) = \alpha_i$ for $i = 1, 2, \dots, 5$. Then we have that

$$E_0(u) = (u(q_0) - u(q_1))^2 + (u(q_0) - u(q_2))^2 + (u(q_0) - u(q_3))^2 + (u(q_0) - u(q_4))^2.$$

We are going to exploit symmetry of the Vicsek set. Since all four "limbs" of the Vicsek set are identical we will only look at one of the limbs since the other limbs will be the same computations. So if we want to minimize $E_1(u')$, we see for the limb between q_0 and q_1 that

$$E_1(u') = (u'(q_1) - u'(p_0))^2 + (u'(p_0) - u'(p_2))^2 + (u'(p_0) - u'(p_3))^2 + (u'(p_0) - u'(p_4))^2 + (u'(p_3) - u'(q_0))^2$$

We are then going to optimize with regards to $u'(p_i)$ for $i = 0, 1, 2, 3$. Taking the derivative of the right side and solving for 0 we get the following

$$\begin{aligned} 4u'(p_0) &= u'(q_1) + u'(p_2) + u'(p_3) + u'(p_4) \\ 2u'(p_3) &= u'(q_0) + u'(p_0) \\ u'(p_2) &= u'(p_0) \\ u'(p_4) &= u'(p_0). \end{aligned}$$

Solving these equations we get

$$\begin{aligned} u'(p_0) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_2) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_4) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_3) &= \frac{2}{3}u'(q_0) + \frac{1}{3}u'(q_1). \end{aligned}$$

So what we have found is that to minimize the energy of the Vicsek set, then the energy of the next point is going to be 2/3 the energy from the closest point and 1/3 the energy from the next closest point.

If one calculates the energies then you will get

$$E_1(u') = \frac{1}{3} \sum_{i=1}^4 (u'(q_i) - u'(q_0))^2 = \frac{1}{3} E_0(u')$$

Since the Vicsek set is constructed from V_m to V_{m+1} by creating a "plus" between any two points that are neighbours in V_m . So going from V_1 to V_2 is the same as going from V_m to V_{m+1} . In which case

$$E_{m+1}(u') = \frac{1}{3} E_m(u')$$

Lemma 1.7: Let $v : K \rightarrow \mathbb{R}$ be a harmonic function on the Vicsek set. If $v(x) = v(y_i) = 0$ for all $y_i \in V_{m,x}$ then $v(\omega) = 0$ for all $\omega \in \overline{B_x}$ where $B_x = \left(\bigcup_{|w|>0} F_w x \right) \cup \left(\bigcup_i \{y_i\} \right)$

Proof. It is clear that $v(\omega) = 0$ for any $\omega \in B_x$, since the energy of the points $F_w x$ with $|w| > 0$ depends on the energy of x and its neighbours.

So we need to check the points in $\overline{B_x} \setminus B_x$. Now since B_x is dense in $\overline{B_x}$, then for any $\omega \in \overline{B_x}$ we have that there exists a sequence $x_n \in B_x$ such that $x_n \rightarrow \omega$. By the fact that v is continuous on K it is continuous on $\overline{B_x}$, in which case $|v(x_n) - v(\omega)| \rightarrow 0$ as $n \rightarrow \infty$, but $v(x_n) = 0$ for all x_n , so $v(\omega) = 0$ as desired. $\square$

What lemma 1.7 states, is if any two points in V_m are neighbours, say x and y , and $v(x) = v(y) = 0$, then the function value at any point between x and y is also zero.

Lemma 1.8: Set $F_m = \text{supp } \psi_x^{(m)}$, then

$$F_0 \supset F_1 \supset F_2 \supset F_3 \supset \dots$$

Further F_m is contained in the open ball $B(x, \epsilon_m)$ where $\epsilon_m \rightarrow 0$ as $m \rightarrow \infty$.

Proof. If we set $\Omega_m = \left\{ z \in K : \psi_x^{(m)}(z) = \psi_x^{(m)}(y) = 0, \forall y \in V_{m,z} \right\}$ then $F_m = K \setminus \left(\bigcup_{\alpha \in \Omega_m} \overline{B_\alpha} \right)$. It is clear that $\Omega_m \subset \Omega_{m+1}$ and that there are new points in Ω_{m+1} that are not just a point between some already existing points $x_1, x_2 \in \Omega_m$. We therefore have from the way that F_m is constructed that $F_m \supset F_{m+1}$.

The containment of F_m in $B(x, \epsilon_m)$ follows from lemma 1.7 and that $F_m \supset F_{m+1}$. $\square$

Theorem 1.9: Let $u \in \text{dom } \Delta_\mu$. Then the pointwise formula

$$\Delta_\mu u(x) = \lim_{m \rightarrow \infty} r^m \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \delta_m u(x)$$

holds with the limit uniform across $V_* \setminus V_0$. Conversely, suppose u is a continuous function and the right side of the limit converges uniformly to a continuous function on $V_* \setminus V_0$. Then $u \in \text{dom } \Delta_\mu$ and the equation holds.

Proof. Suppose $u \in \text{dom } \Delta_\mu$ and $\Delta_\mu u = f$, then by [\[INSERT REF\]](#) we have that

$$r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \Delta_m u(x) = \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu}$$

for any $x \in V_m \setminus V_0$. Uniform converges of the right side is because of continuity of $f(x)$, so let $\epsilon > 0$ be given, then by lemma 1.8 we can choose m large enough so for all $x, y \in F_m$ that $|f(x) - f(y)| < \epsilon$, then

$$\begin{aligned}
 \left| \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} - f(x) \right| &= \left| \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} - \frac{\int_K f(x) \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} \right| \\
 &= \left| \frac{\int_K (f(y) - f(x)) \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} \right| \\
 &\leq \frac{\int_{F_m} |f(y) - f(x)| d\mu}{\int_K \psi_x^{(m)} d\mu} \\
 &\leq \frac{\epsilon \int_{F_m} d\mu}{\int_K \psi_x^{(m)} d\mu} \\
 &= \epsilon
 \end{aligned}$$

in which case

$$\lim_{m \rightarrow \infty} r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \Delta_m u(x) = \lim_{m \rightarrow \infty} \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} = f(x) = \Delta_\mu u(x)$$

Moving on to the second half of the proof. Suppose that $r^m \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \delta_m u(x)$ converges uniformly to a continuous function f . Now for any $v \in \text{dom}_0 \mathcal{E}$ we have

$$\mathcal{E}_m(u, v) = r^{-m} \bigotimes_{\substack{x \sim y \\ m}} (u(y) - u(x)) (v(y) - v(x)).$$

If we collect all the terms that contain the factor $v(x)$ for a fixed $x \in V_m \setminus V_0$ then each $v(x)$ will be of the form $-v(x) (u(y) - u(x))$ for each $y \sim_m x$. In which case we get

$$\begin{aligned}
 \mathcal{E}_m(u, v) &= -r^{-m} \bigotimes_{x \in V_m \setminus V_0} v(x) \left(\bigotimes_{\substack{x \sim y \\ m}} (u(y) - u(x)) \right) \\
 &= - \bigotimes_{x \in V_m \setminus V_0} v(x) r^{-m} \Delta_m u(x)
 \end{aligned}$$

While it seems we lose many terms by removing $v(y)$, but they are instead contained in $\bigotimes_{x \sim_m y} (u(y) - u(x))$. Note that the sum is "reset" for every x so we count both the relation $x \sim y$ and $y \sim x$.

Now if we define $f_m(x) = r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \Delta_m u(x)$ then $f_m \rightarrow f$ uniformly by assumption and we can rewrite the above as

$$\begin{aligned}
 \mathcal{E}_m(u, v) &= - \bigotimes_{x \in V_m \setminus V_0} v(x) \left(\int_K \psi_x^{(m)} d\mu \right) r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \Delta_m u(x) \\
 &= - \int_K \left(\bigotimes_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)} \right) d\mu.
 \end{aligned}$$

Following a quite lengthy argument, we have that

$$\mathcal{E}(u, v) = \lim_{m \rightarrow \infty} \mathcal{E}_m(u, v) = \lim_{m \rightarrow \infty} - \int_K \left(\bigotimes_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)} \right) d\mu = - \int_K v(y) f(y) d\mu.$$

so $u \in \text{dom}_\mu$

$$\begin{aligned} & \left| \bigotimes_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) - v(y) f(y) \right| \\ & \leq \left| \bigotimes_{x \in V_m \setminus V_0} (v(x) f_m(x) - v(y) f(y)) \psi_x^{(m)}(y) \right| + \left| \bigotimes_{x \in V_m \setminus V_0} v(y) f(y) \psi_x^{(m)}(y) - v(y) f(y) \right| \end{aligned}$$

Looking at each term individually we see that

$$\begin{aligned} & \left| \bigotimes_{x \in V_m \setminus V_0} (v(x) f_m(x) - v(y) f(y)) \psi_x^{(m)}(y) \right| \\ & \leq \bigotimes_{x \in V_m \setminus V_0} \left(|v(x) f_m(x) - v(y) f_m(y)| + |v(y) f_m(y) - v(y) f(y)| \right) \psi_x^{(m)}(y) \end{aligned}$$

now again we will look at each term starting with $|v(y) f_m(y) - v(y) f(y)|$. Here

$$|v(y) f_m(y) - v(y) f(y)| = |v(y)| |f_m(y) - f(y)|$$

and since v is continuous on K , which is compact, there is a constant C_1 so $|v(y)| \leq C_1$ for all $x \in K$. Since f_m converges uniformly to f we can choose m_1 such that $|f_m(y) - f(y)| \leq \frac{\epsilon}{C_1}$ and the convergence of that term follows.

Next is $|v(x) f_m(x) - v(y) f_m(y)|$, here we find

$$|v(x) f_m(x) - v(y) f_m(y)| \leq |v(x)| |f_m(x) - f(y)| + |f_m(y)| |v(x) - v(y)|$$

First $|f_m(y)| |v(x) - v(y)|$, here $|f_m(y)| \leq C_2$ is uniformly bounded by some C_2 **why?**. By lemma 1.8 and continuity of v we can choose m_2 so $|v(x) - v(y)| \leq \frac{\epsilon}{C_2}$ for all $x \in F_{m_2}$. The reason we can choose m_2 is because $\psi_x^{(m)}$ is multiplied onto all the terms so we just restrict v to F_{m_2} .

Second $|v(x)| |f_m(x) - f(y)|$, again $|v(x)| \leq C_1$, but

$$|f_m(x) - f(y)| \leq |f_m(x) - f(x)| + |f(x) - f_m(y)| \leq |f_m(x) - f(x)| + |f(x) - f(y)| + |f(y) - f_m(y)|$$

First choose m_3 so $|f_{m_3}(x) - f(x)| \leq \frac{\epsilon}{3C_1}$, next by continuity of f choose m_4 so $|f(x) - f(y)| \leq \frac{\epsilon}{3C_1}$ for all $x \in F_{m_4}$, lastly choose m_5 so $|f(y) - f_{m_5}(y)| \leq \frac{\epsilon}{3C_1}$.

So we now have for $m \geq \max\{m_1, m_2, \dots, m_5\}$ that

$$\left| \bigotimes_{x \in V_m \setminus V_0} (v(x) f_m(x) - v(y) f(y)) \psi_x^{(m)}(y) \right| \leq \epsilon \bigotimes_{x \in V_m \setminus V_0} \psi_x^{(m)} \leq \epsilon \bigotimes_{x \in V_m \setminus V_0} 1 \leq \epsilon.$$

We are now going to look at $\bigotimes_{x \in V_m \setminus V_0} v(y) f(y) \psi_x^{(m)}(y) - v(y) f(y)$ here

$$\left| \bigotimes_{x \in V_m \setminus V_0} v(y) f(y) \psi_x^{(m)}(y) - v(y) f(y) \right| = |v(y) f(y)| \left| \bigotimes_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right|$$

Now to show $\left| \bigotimes_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right| = 0$ as $m \rightarrow \infty$, we start by looking at $y \in V^*$. Choose m_6 so $y \in V_{m_6}$.

Next assume $y \in K$, then since V^* is dense in K we can choose a sequence $(x_n)_{n=1}$ where $x_n \rightarrow y$ as $n \rightarrow \infty$, and since $\psi_x^{(m)}$ is continuous on K then $\psi_x^{(m)}(x_n) \rightarrow \psi_x^{(m)}(y)$ as $n \rightarrow \infty$. This gives us the following

$$\left| \bigotimes_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right| \leq \left| \bigotimes_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - \bigotimes_{x \in V_m \setminus V_0} \psi_x^{(m)}(x_n) \right| + \left| \bigotimes_{x \in V_m \setminus V_0} \psi_x^{(m)}(x_n) - 1 \right|$$

Now choose an n_1 so $\left| \psi_x^{(m)}(x_{n_1}) - \psi_x^{(m)}(y) \right| \leq \epsilon$ and an m_7 so $x_{n_1} \in V_{m_7}$. Both can then be bounded by $m \geq \max\{n_1, m_7\}$ $\square$

Chapter 2

And now for something completely different

In the previous chapter we constructed the Vicsek set, now we will consider the discrete Laplacian on this

Definition 2.1: Let $\ell(V_m)$ be the set of functions of $f : V_m \rightarrow \mathbb{R}$. Then for any $f \in \ell(V_m)$ the *discrete Laplacian* on V_m is defined as

$$\Delta_m f(p) = \text{CONST} \sum_{q \in V_{m,p}} (f(q) - f(p)), \quad p \in V_m \setminus V_0$$

Now let $C(K)$ be the set of continuous functions on K . We then define

$$\mathcal{D} = \left\{ f \in C(K) : \exists g \in C(K), \lim_{m \rightarrow \infty} \max_{p \in V_m \setminus V_0} |\Delta_m f(p) - g(p)| = 0 \right\}$$

This set leads us to defining

Definition 2.2: The Kigami Laplacian $\Delta : \mathcal{D} \rightarrow C(K)$ is defined as

$$\Delta f(p) = g(p)$$

where g satisfies: $\lim_{m \rightarrow \infty} \max_{p \in V_m \setminus V_0} |\Delta_m f(p) - g(p)| = 0$

Theorem 2.3 (Bochner-Minlos): Assume $\mathbb{X}$ is a nuclear space. Any continuous positive-definite linear functional Λ on $\mathbb{X}$ satisfying $\Lambda(0) = 1$ is the Fourier transform of a countably additive positive normalized measure on $\mathbb{X}$ **REFERENCE IS FROM WIKIPEDIA** Ref is here

Theorem 2.4: Let $s \geq 0$, there exists a centered Gaussian distribution X_s on $S'(K)$ such that for $f, g \in S(K)$

$$\mathbb{E}[X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu$$

Proof. The space $S(K)$ is a nuclear space **WHY**, it is enough to use Bochner-Minlos theorem 2.3 on the functional

$$\Lambda : f \rightarrow \exp \left(-\frac{1}{2} \|f\|_k^2 \right).$$

DEFINE NORM

This is enough since theorem 2.3 then gives an unique probability measure ν such

$$\exp \left(-\frac{1}{2} \|f\|_k^2 \right) = \int_{S'(K)} e^{i\langle x, f \rangle} \, d\nu(x).$$

Notice that the right side is an characteristic function for some random variable. If we choose $f = t \cdot g$ for some $g \in S(K)$ and $t \in \mathbb{R}$ **Where is t from?** (notice that $t \cdot g \in S(K)$) then the left side of the equation becomes the characteristic function for the normal distribution with mean zero and variance $\|f\|_s^2$. Thus we have the X_s we wanted.

Now to prove that Λ satisfies the requirements of theorem 2.3. Obviously $\Lambda(0) = 1$, further it follows from the fact that $\|t \cdot f\|_k^2$ is an inner product that with **THEOREM: 2.4 A SURVEY** Λ is positive definite. Lastly we need to prove it is continuous at zero.

First we note that since $(-\Delta)^{-s}$ is a bounded operator we have that there exists some constant $C > 0$ so $\|(-\Delta)^{-s} f\|_{L^2(K, \mu)} \leq C \|f\|_{L^2(K, \mu)}$.

Now let $\epsilon > 0$ be given, choose

$$\delta = \begin{cases} \left(\frac{-2 \cdot \ln(1-\epsilon)}{C} \right)^{\frac{1}{2}}, & \text{when } \epsilon \in (0, 1) \\ 1, & \text{when } \epsilon \geq 1. \end{cases}$$

First if $\epsilon \geq 1$ notice that

$$\left| \exp \left(-\frac{1}{2} \|f\|_s^2 \right) - \exp \left(-\frac{1}{2} \|0\|_s^2 \right) \right| \leq 1 \leq \epsilon, \quad \forall f \in S(K)$$

in which we can choose δ to be any real number larger than 0. Moving on to $\epsilon \in (0, 1)$ we get

$$\begin{aligned} \left| \exp \left(-\frac{1}{2} \|f\|_s^2 \right) - \exp \left(-\frac{1}{2} \|0\|_s^2 \right) \right| &\leq \left| \exp \left(-\frac{1}{2} C \|f\|_{L^2(K, \mu)}^2 \right) - 1 \right| \\ &= \left| \exp \left(-\frac{1}{2} C \int_K |f|^2 d\mu \right) - 1 \right| \\ &\leq \left| \exp \left(-\frac{1}{2} C \int_K \delta^2 d\mu \right) - 1 \right| \\ &= \left| \exp \left(-\frac{1}{2} C \int_K \frac{-2 \cdot \ln(1-\epsilon)}{C} d\mu \right) - 1 \right| \\ &= \left| \exp (\ln(1-\epsilon)) - 1 \right| \\ &= \epsilon \end{aligned}$$

in the third to last equality using that μ is a probability measure. In which case we have continuity in 0 as desired. $\square$

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